

Safety data sheet according to UK REACH

Printing date 28.05.2026

Version number 3.5 (replaces version 3.4)

Revision: 12.06.2025

* SECTION 1: Identification of the substance/mixture and of the company/undertaking

- **1.1 Product identifier**
- **Trade name:** Tretinoin
- **Chemical Identification:**
(2E,4E,6E,8E)-3,7-dimethyl-9-(2,6,6-trimethylcyclohexen-1-yl)nona-2,4,6,8-tetraenoic acid
- **Article number:** 704148, 704149, 704150
- **EC number:**
206-129-0
- **1.2 Relevant identified uses of the substance or mixture and uses advised against** n/a
- **Application of the substance / the mixture** Active pharmaceutical ingredients
- **1.3 Details of the supplier of the safety data sheet**
- **Manufacturer/Supplier:**
Siegfried AG
Untere Brühlstrasse 4
CH-4800 Zofingen
- **Informing department:**
Siegfried AG
Tel.: +41 62 746 14 78
Fax: +41 62 746 14 01
E-Mail: she@siegfried.ch
- **1.4 Emergency telephone number:**
Members of the public seeking specific information on poisons should contact:
In England and Wales: NHS 111 - dial 111
In Scotland: NHS 24 - dial 111

* SECTION 2: Hazards identification

- **2.1 Classification of the substance or mixture**
- **Classification according to Regulation (EC) No 1272/2008**
Skin Irrit. 2 H315 Causes skin irritation.
Repr. 1B H360D May damage the unborn child.
Aquatic Acute 1 H400 Very toxic to aquatic life.
Aquatic Chronic 1 H410 Very toxic to aquatic life with long lasting effects.
- **2.2 Label elements**
- **Labelling according to Regulation (EC) No 1272/2008**
The substance is classified and labelled according to the GB CLP regulation.
- **Hazard pictograms**


GHS07 GHS08 GHS09
- **Signal word** Danger
- **Hazard statements**
H315 Causes skin irritation.

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H360D May damage the unborn child.

H410 Very toxic to aquatic life with long lasting effects.

Precautionary statements

P273 Avoid release to the environment.

P280 Wear protective gloves/protective clothing/eye protection/face protection/hearing protection.

P308+P313 IF exposed or concerned: Get medical advice/attention.

P332+P313 If skin irritation occurs: Get medical advice/attention.

P405 Store locked up.

P501 Dispose of contents/container in accordance with local/regional/national/international regulations.

SECTION 3: Composition/information on ingredients**3.1 Substances****CAS No. Designation:**

Tretinoin

Identification number(s):**EC number:** 206-129-0**SECTION 4: First aid measures****4.1 Description of first aid measures****General information** Show this safety data sheet to the attending physician.**After inhalation**

Seek medical treatment in case of complaints.

In case of unconsciousness bring patient into stable side position for transport.

After skin contact

If skin irritation continues, consult a doctor.

Instantly wash with water and soap and rinse thoroughly.

After eye contact Rinse opened eye for several minutes under running water.**After swallowing** Rinse out mouth and then drink plenty of water.**4.2 Most important symptoms and effects, both acute and delayed**

No further relevant information available.

4.3 Indication of any immediate medical attention and special treatment needed

No further relevant information available.

*** SECTION 5: Firefighting measures****5.1 Extinguishing media****Suitable extinguishing agents**CO₂, extinguishing powder or water jet. Fight larger fires with water jet or alcohol-resistant foam.**5.2 Special hazards arising from the substance or mixture** Carbon oxides (CO_x)**5.3 Advice for firefighters****Protective equipment:** Wear self-contained breathing apparatus.

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SECTION 6: Accidental release measures

· 6.1 Personal precautions, protective equipment and emergency procedures

Use breathing protection against the effects of fumes/dust/aerosol.

Avoid causing dust.

Wear protective equipment. Keep unprotected persons away.

· 6.2 Environmental precautions:

Inform respective authorities in case product reaches water or sewage system.

Damp down dust with water spray jet.

Do not allow to enter drainage system, surface or ground water.

Keep dirty washing water for appropriate disposal.

· 6.3 Methods and material for containment and cleaning up:

Dispose of contaminated material as waste according to section 13.

· 6.4 Reference to other sections

See Section 7 for information on safe handling

See Section 8 for information on personal protection equipment.

See Section 13 for information on disposal.

* SECTION 7: Handling and storage

· 7.1 Precautions for safe handling

Provide suction extractors if dust is formed.

Keep containers tightly sealed.

· Information about protection against explosions and fires: No special measures required.

· 7.2 Conditions for safe storage, including any incompatibilities

· Storage

· Requirements to be met by storerooms and containers: Store only in the original container.

· Information about storage in one common storage facility: Store away from foodstuffs.

· Further information about storage conditions: Protect from the effects of light.

· 7.3 Specific end use(s) No further relevant information available.

SECTION 8: Exposure controls/personal protection

· 8.1 Control parameters

· Components with limit values that require monitoring at the workplace: Not required.

· Additional information: The lists that were valid during the compilation were used as basis.

· Exposure category (Siegfried) 3 (OEL-Value $\mu\text{g}/\text{m}^3 < 10$; ≥ 1)

· 8.2 Exposure controls

· Appropriate engineering controls No further data; see section 7.

· Individual protection measures, such as personal protective equipment

· General protective and hygienic measures

Pregnant women must strictly avoid inhalation or contact with the skin.

Keep away from foodstuffs, beverages and food.

Take off immediately all contaminated clothing

Wash hands during breaks and at the end of the work.

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Avoid contact with the skin.

Avoid contact with the eyes and skin.

· **Breathing equipment:** Filter P3.

· **Hand protection**

Protective gloves.

The glove material has to be impermeable and resistant to the product/ the substance/ the preparation.

Due to missing tests no recommendation to the glove material can be given for the product/ the preparation/ the chemical mixture.

Selection of the glove material on consideration of the penetration times, rates of diffusion and the degradation

· **Material of gloves**

The selection of the suitable gloves does not only depend on the material, but also on further marks of quality and varies from manufacturer to manufacturer.

· **Penetration time of glove material**

The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.

· **Eye/face protection** Tightly sealed safety glasses.

· **Body protection:** Protective work clothing.

SECTION 9: Physical and chemical properties

· **9.1 Information on basic physical and chemical properties**

· **General Information**

· **Physical state**

Solid.

· **Colour:**

Orange

· **Odour:**

Odourless

· **Odour threshold:**

Not determined.

· **Melting point/freezing point:**

176 – 182 °C

· **Boiling point or initial boiling point and boiling range**

Not determined

· **Flammability**

Product is not inflammable.

· **Burning test**

· **Lower and upper explosion limit**

· **Lower:**

Not determined.

· **Upper:**

Not determined.

· **Flash point:**

Not applicable

· **Auto-ignition temperature:**

Not determined.

· **Decomposition temperature:**

Not determined.

· **pH**

8.2 (0.01%)

· **Viscosity:**

· **Kinematic viscosity**

Not applicable.

· **dynamic:**

Not applicable.

· **Solubility**

· **Water:**

Not determined.

· **Partition coefficient n-octanol/water (log value)**

0.82607

· **Vapour pressure:**

Not applicable.

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· Density and/or relative density	
· Density	Not determined
· Relative density	Not determined.
· Settled apparent density	480 kg/m ³
· Vapour density	Not applicable.
· 9.2 Other information	
· Appearance:	
· Form:	Crystalline powder
· Important information on protection of health and environment, and on safety.	
· Self-inflammability:	Not determined.
· Explosive properties:	Product is not explosive.
· Change in condition	
· Evaporation rate	Not applicable.
· Information with regard to physical hazard classes	
· Explosives	Void
· Flammable gases	Void
· Aerosols	Void
· Oxidising gases	Void
· Gases under pressure	Void
· Flammable liquids	Void
· Flammable solids	Void
· Self-reactive substances and mixtures	Void
· Pyrophoric liquids	Void
· Pyrophoric solids	Void
· Self-heating substances and mixtures	Void
· Substances and mixtures, which emit flammable gases in contact with water	Void
· Oxidising liquids	Void
· Oxidising solids	Void
· Organic peroxides	Void
· Corrosive to metals	Void
· Desensitised explosives	Void

SECTION 10: Stability and reactivity

- **10.1 Reactivity** No further relevant information available.
- **10.2 Chemical stability**
- **Thermal decomposition / conditions to be avoided:** No decomposition if used according to specifications.
- **10.3 Possibility of hazardous reactions** No dangerous reactions known
- **10.4 Conditions to avoid** No further relevant information available.
- **10.5 Incompatible materials:** No further relevant information available.
- **10.6 Hazardous decomposition products:** No dangerous decomposition products known

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SECTION 11: Toxicological information

- **11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008**
- **Acute toxicity** Based on available data, the classification criteria are not met.
- **LD/LC50 values that are relevant for classification:**
Oral LD50 8000 mg/kg (rat)
- **Primary irritant effect:**
- **Skin corrosion/irritation** Causes skin irritation.
- **Reproductive toxicity** May damage the unborn child.
- **11.2 Information on other hazards**
- **Endocrine disrupting properties** Substance is not listed.

SECTION 12: Ecological information

- **12.1 Toxicity**
- **Aquatic toxicity:**
LC50 0.2 mg/l /96 h (Fish) (OECD 203)
EC50 (static) > 0.1 – < 0.3 mg/l /4 (daphne) (OECD 202)
- **12.2 Persistence and degradability**
EC20 > 1000 mg/l /0.5 (activated sludge) (OECD 209)
- **12.3 Bioaccumulative potential** No further relevant information available.
- **12.4 Mobility in soil** No further relevant information available.
- **12.5 Results of PBT and vPvB assessment**
- **PBT:** This substance does not meet PBT criteria of Regulation (EC) No 1907/2006, Annex XIII.
- **vPvB:** This substance does not meet vPvB criteria of Regulation (EC) No 1907/2006, Annex XIII.
- **12.6 Endocrine disrupting properties**
The product does not contain substances with endocrine disrupting properties.
- **12.7 Other adverse effects**
- **Remark:** Very toxic for fish
- **Additional ecological information:**
- **General notes:**
Water danger class 3 (German Regulation) (Self-assessment): extremely hazardous for water.
Do not allow product to reach ground water, water bodies or sewage system, even in small quantities.
Danger to drinking water if even extremely small quantities leak into soil.
Also poisonous for fish and plankton in water bodies.
Very toxic for aquatic organisms

SECTION 13: Disposal considerations

- **13.1 Waste treatment methods**
- **Recommendation**
Must not be disposed of together with household garbage. Do not allow product to reach sewage system.

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- **Uncleaned packagings:**
- **Recommendation:** Disposal must be made according to official regulations.

SECTION 14: Transport information

- **14.1 UN number or ID number** UN3077
- **ADR/RID, IMDG, IATA**
- **14.2 UN proper shipping name** ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (Acide rétinoïque)
- **ADR/RID** ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (Tretinoin), MARINE POLLUTANT
- **IMDG** Environmentally hazardous substance, solid, n.o.s. (Tretinoin)
- **IATA**
- **14.3 Transport hazard class(es)**
- **ADR/RID, IMDG, IATA**
- 
- **Class** 9 Miscellaneous dangerous substances and articles.
- **Label** 9
- **14.4 Packing group**
- **ADR/RID, IMDG, IATA** III
- **14.5 Environmental hazards:**
- **Marine pollutant:** Symbol (fish and tree)
- **Special marking (ADR/RID):** Symbol (fish and tree)
- **Special marking (IATA):** Symbol (fish and tree)
- **14.6 Special precautions for user** Warning: Miscellaneous dangerous substances and articles.
- **Kemler Number:** 90
- **EMS Number:** F-A,S-F
- **Stowage Category** A
- **Stowage Code** SW23 When transported in BK3 bulk container, see 7.6.2.12 and 7.7.3.9.
- **14.7 Maritime transport in bulk according to IMO instruments** Not evaluated
- **Transport/Additional information:**
- **ADR/RID**
- **Limited quantities (LQ)** 5 kg
- **Excepted quantities (EQ)** Code: E1
Maximum net quantity per inner packaging: 30 g
Maximum net quantity per outer packaging: 1000 g
- **Transport category** 3

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- **Tunnel restriction code** (-)
- **IMDG**
- **Limited quantities (LQ)** 5 kg
- **Excepted quantities (EQ)** Code: E1
Maximum net quantity per inner packaging: 30 g
Maximum net quantity per outer packaging: 1000 g
- **UN "Model Regulation":** UN 3077 ENVIRONMENTALLY HAZARDOUS
SUBSTANCE, SOLID, N.O.S. (ACIDE RÉTINOIQUE), 9,
III

SECTION 15: Regulatory information

- **15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture**
- **Directive 2012/18/EU**
- **Named dangerous substances - ANNEX I** Substance is not listed.
- **Seveso category** E1 Hazardous to the Aquatic Environment
- **Qualifying quantity (tonnes) for the application of lower-tier requirements** 100 t
- **Qualifying quantity (tonnes) for the application of upper-tier requirements** 200 t
- **National regulations**
- **Water hazard class:** Water danger class 3 (Self-assessment): extremely hazardous for water.
- **15.2 Chemical safety assessment:** A Chemical Safety Assessment has been carried out.

* SECTION 16: Other information

These data are based on our present knowledge. However, they shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

- **Abbreviations and acronyms:**
ADR: Accord relatif au transport international des marchandises dangereuses par route (European Agreement Concerning the International Carriage of Dangerous Goods by Road)
IMDG: International Maritime Code for Dangerous Goods
IATA: International Air Transport Association
GHS: Globally Harmonised System of Classification and Labelling of Chemicals
EINECS: European Inventory of Existing Commercial Chemical Substances
CAS: Chemical Abstracts Service (division of the American Chemical Society)
LC50: Lethal concentration, 50 percent
LD50: Lethal dose, 50 percent
PBT: Persistent, Bioaccumulative and Toxic
vPvB: very Persistent and very Bioaccumulative
Skin Irrit. 2: Skin corrosion/irritation – Category 2
Repr. 1B: Reproductive toxicity – Category 1B
Aquatic Acute 1: Hazardous to the aquatic environment - acute aquatic hazard – Category 1
Aquatic Chronic 1: Hazardous to the aquatic environment - long-term aquatic hazard – Category 1
- *** Data compared to the previous version altered.**